I 320D · Data Science for Biomedical Informatics · Spring 2026

Step 0: Environment Setup & Data Download

Expanded Reference · Challenge Questions with Full Answers

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What this document contains

This cheatsheet expands Step 0 of the Project 4 notebook into a complete standalone reference. It covers:

- Full step-by-step instructions for environment setup, data download, and integrity checks
- Complete model answers to all four Step 0 challenge questions, written at graduate level
- Side-by-side comparison tables, decision frameworks, and code snippets ready to use
- A self-assessment checklist so you can verify your understanding before the lab session

All four challenge question answers include: the reasoning chain, supporting evidence, code where applicable, and common pitfalls to avoid.

PART 1 — Step-by-Step Instructions

1. Establish a reproducible Python environment with all dependencies pinned

A reproducible environment guarantees that every student, TA, and grader produces identical results. Follow this sequence:

- **Create a virtual environment**

```
python -m venv venv && source venv/bin/activate   # Linux/Mac
python -m venv venv && venv\Scripts\activate      # Windows
```

- **Install all required packages**

```
pip install pandas numpy matplotlib seaborn scikit-learn torch tqdm shap nbformat
```

- **Pin all versions immediately**

```
pip freeze > requirements.txt
```

- **Fix random seeds at top of notebook**

```
import numpy as np, torch
np.random.seed(42)
torch.manual_seed(42)
```

- **Commit environment file alongside notebook**

```
git add requirements.txt notebook.ipynb && git commit -m 'pin environment'
```

> ⚠ **Why this matters in clinical AI**
>
> The FDA's guidance on AI/ML-based Software as a Medical Device requires full documentation of the software environment. A paper where results cannot be reproduced is a liability, not just a quality issue. Reviewers at top clinical informatics venues (JAMIA, JBI, Nature Digital Medicine) now require requirements.txt or Docker images alongside submissions.

2. Download both PhysioNet 2019 training sets (Set A = BIDMC, Set B = Emory)

The notebook handles downloads programmatically. If running manually:

```
Set A URL:  https://archive.physionet.org/users/shared/challenge-2019/training_setA.zip
Set B URL:  https://archive.physionet.org/users/shared/challenge-2019/training_setB.zip
Expected sizes:  Set A ≈ 28 MB  |  Set B ≈ 23 MB
Expected file counts:  Set A = 20,336 .psv files  |  Set B = 20,000 .psv files
```

Why two sets? Training and evaluating on patients from two different hospitals provides a more honest estimate of cross-site generalisation — the same challenge a model faces when it is deployed at a new institution. If you train and test on the same hospital, you might learn hospital-specific patterns (equipment calibration, documentation habits) rather than universal sepsis physiology.

3. Understand the physical file layout: one .psv file per patient, rows are hourly ICU measurements

Column group	Examples	Typical missingness	Update frequency

Vitals (8 cols)	HR, O2Sat, Temp, SBP, MAP, DBP, Resp, EtCO2	5 – 30%	Every 1 – 5 min (aggregated to 1 hr)
Labs (26 cols)	Lactate, Creatinine, WBC, HCO3, pH, Glucose	40 – 95%	1 – 3× per day at most
Demographics (6 cols)	Age, Gender, Unit1, Unit2, HospAdmTime, ICULOS	< 2%	Constant within stay
Label (1 col)	SepsisLabel	0%	0 until onset, then 1 forever

Quick inspection command — run this after loading one file to verify structure:

```
import pandas as pd
df = pd.read_csv('physionet2019/training/training_setA/p000001.psv', sep='|')
print(df.shape, df.columns.tolist())  # expect (N_hours, 41)
print(df.isnull().mean().sort_values(ascending=False).head(10))  # missingness check
```

4. Confirm data integrity by checking file counts and directory structure

Run this integrity check block immediately after extraction:

```
import os, glob
for folder, expected in [('training_setA', 20336), ('training_setB', 20000)]:
    path = f'physionet2019/training/{folder}'
    n    = len(glob.glob(f'{path}/*.psv'))
    ok   = '✓' if n == expected else '✗ MISMATCH'
    print(f'{ok}  {folder}: {n} files (expected {expected})')
```

If counts are off: re-run the download cell (partial download is common on slow connections). Do NOT proceed to Step 1 with incomplete data — the glob+concat loop will silently produce a smaller dataset and your AUROC will be unrepresentative.

PART 2 — Challenge Questions: Full Answers

Q1 PhysioNet & CITI credentialing: what makes the Challenge dataset different from MIMIC?

HINT *Look at the Challenge's data description page on PhysioNet. What identifiers were removed? Were dates shifted? Are ages exact or top-coded at 90?*

✓ THE SHORT ANSWER

The 2019 Challenge dataset was released under a less restrictive access policy because it underwent more aggressive de-identification than MIMIC. Specifically: it contains no individually identifiable information (dates, geographic identifiers, exact ages for elderly patients), was prepared specifically for a public machine learning competition, and was reviewed and released under PhysioNet's open-access credential tier rather than the credentialed tier.

✓ SIDE-BY-SIDE COMPARISON

MIMIC-III / MIMIC-IV (Credentialed — requires CITI)	PhysioNet 2019 Challenge (Open access — no CITI)
Contains real hospital dates (shifted by random offset per patient but self-consistent)	All dates removed — only relative time offsets remain (ICULOS = hours since ICU admission)
Ages exact up to 89; patients ≥ 90 are top-coded as 91.4 to prevent re-identification	Ages not publicly documented as top-coded; challenge organisers removed direct age identifiers
Contains geographic information (hospital IDs, some state-level data)	Hospital identity partially disclosed (BIDMC, Emory) but no patient-level geography
Contains free-text clinical notes (rich but highly re-identifiable)	No free text — only structured numeric measurements
Contains ICD billing codes (disease categories that could narrow identity)	No diagnosis codes included
Requires CITI certification (~3 hrs) + PhysioNet credentialed account + DUA	Freely downloadable ZIP files, no account required
IRB oversight: MIT and Beth Israel Deaconess Medical Center	Challenge data prepared under challenge organisers' IRB; no ongoing DUA for users

✓ HIPAA SAFE HARBOR: THE 18 IDENTIFIERS

HIPAA's Safe Harbor method (45 CFR §164.514(b)) requires removing 18 categories of identifiers. MIMIC applies Safe Harbor with date shifting. The 2019 Challenge dataset eliminates or never collected most of these:

Identifier	MIMIC treatment	Challenge treatment	Re-ID risk if present
Names	Removed	Never collected	High
Geographic data (zip, state)	State-level retained in some tables	Removed	Medium
Dates	Shifted by random per-patient offset	Replaced with relative hours (ICULOS)	High without shift
Ages ≥ 90	Top-coded as 91.4 years	Not documented; challenge removed exact birth dates	Medium
Phone / fax numbers	Removed	Never collected	Low in ICU context
Device identifiers (e.g. pacemaker IDs)	Removed	Never collected	Medium
IP addresses	Removed	Never collected	Low
Full-face photos	Removed	Never collected	N/A
Account / medical record numbers	Replaced with random ID	Random patient file name (p000001.psv)	Medium without replacement

✓ THE DEEPER ANSWER: WHAT MAKES A DATASET OPEN-ACCESS?

Three factors determine whether a clinical dataset can be released without individual user credentialing:

Aggregation risk: Can you cross-reference it with other public data (social media, obituaries, news) to re-identify individuals? MIMIC with shifted dates is harder to cross-reference but still possible — hence credentialing. The Challenge data with no dates at all is far harder to cross-reference.

Consent and IRB scope: MIMIC patients consented under an IRB protocol specifying credentialed access. The Challenge organisers had a different consent and IRB scope that permitted open release.

Intended use: MIMIC is a general research database used for diverse studies including potentially sensitive ones (mental health, addiction). The Challenge data was scoped to a specific ML competition use case with more restricted scope.

Key exam takeaway

The critical distinction is not just which 18 identifiers were removed, but the combination of: (1) the de-identification method applied, (2) the aggregation risk given the remaining data, and (3) the IRB/consent framework under which the data was collected. MIMIC requires CITI because it is rich enough to potentially re-identify patients even after de-identification; the Challenge dataset is not.

Q2 Automating hourly data ingestion from a live EHR for production deployment

HINT *Think HL7 FHIR R4 APIs, HL7 v2 feeds, or direct database connections. What is the typical lag between a nurse charting a vital sign and it appearing in the database?*

✓ THE SHORT ANSWER

Two industry-standard interfaces dominate clinical data integration: (1) HL7 FHIR R4 REST APIs for modern EHR platforms, and (2) HL7 v2 message feeds for legacy systems. A third option — direct database queries — is common in academic medical centres but has significant governance concerns. Expected end-to-end latency ranges from 2 to 15 minutes for most systems, meaning an hourly prediction schedule is realistic.

✓ INTERFACE COMPARISON TABLE

Interface	How it works	Typical latency	Best for
HL7 FHIR R4 REST API	HTTP GET/POST to standardised resource endpoints (Observation, Patient, Encounter). Every EHR vendor must support since 2021 ONC rule.	5 – 15 min (poll-based); < 1 min with FHIR Subscriptions (push)	Epic, Cerner, Oracle Health — modern EHRs
HL7 v2 ADT / ORU feeds	Pipe-delimited message streams. ADT messages = admissions/transfers/discharges. ORU messages = lab results. Delivered via MLLP TCP socket.	1 – 5 min (near-real-time push)	Legacy systems, labs, older hospital middleware
Direct DB query (SQL)	Read from EHR's backend database (e.g. Epic Clarity, Cerner Millennium DB) via read-replica. Requires DBA access and strict governance.	< 1 min (near-real-time with triggers)	Research environments with direct DB agreements
REDCap / data warehouse API	Query from enterprise data warehouse (EDW) that aggregates EHR data. E.g. Epic's Chronicles or Caboodle.	15 – 60 min (EDW ETL lag)	Research queries, not real-time alerting

✓ LATENCY BREAKDOWN: WHERE THE TIME GOES

When a nurse charts a vital sign, the data travels through multiple systems before your model can use it:

Stage	Typical lag	Notes

Nurse charts vital in EHR	**0 – 2 min**	Depends on charting workflow; some units chart retrospectively
EHR commits to production DB	**< 30 sec**	Near-instant for modern EHRs
HL7 message generated and transmitted	**< 1 min**	ADT/ORU messages sent on DB commit trigger
Integration engine (Mirth, Rhapsody) routes message	**1 – 5 min**	Queue depth depends on hospital volume
API endpoint indexes new observation	**1 – 10 min**	FHIR server indexing lag; varies by vendor
Your ingestion script polls or receives push	**Polling: 0 – 60 min cycle; Push: < 1 min**	Subscriptions eliminate most of this lag
Feature engineering + model inference	**< 1 min**	Assuming pre-computed rolling features cached
TOTAL end-to-end	**5 – 75 min**	Polling-based; 3 – 10 min with push subscriptions

✓ MINIMAL PRODUCTION ARCHITECTURE

A minimal viable production pipeline for hourly sepsis scoring:

```
EHR (Epic/Cerner)
  ↓  HL7 v2 ORU/ADT or FHIR R4 Subscription
Integration Engine (Mirth Connect / Azure APIM)
  ↓  Normalised message to canonical schema
Streaming buffer (Kafka topic or AWS Kinesis)
  ↓  Consumer reads every 5 min (or on push)
Feature service (FastAPI + Redis cache)
  ↓  Computes rolling stats, delta features per patient
Model inference (ONNX Runtime or TorchServe)
  ↓  Outputs P(sepsis) per patient-hour
Alert router (PagerDuty / EHR Best Practice Advisory)
  ↓  Fires alert to nurse workstation if P > threshold
```

⚠ Critical compliance requirements for production deployment

HIPAA BAA required with every cloud vendor touching PHI (AWS, Azure, GCP all offer this)

ONC 21st Century Cures Act: EHR vendors must provide FHIR R4 access — you have legal leverage to request it

Hospital IRB approval required before your model touches live patient data, even in read-only mode

FDA SaMD (Software as a Medical Device) pathway may apply if the alert drives clinical decisions

Model must be validated on the specific hospital's patient population before go-live

Q3 Learning curves: how AUROC changes with training cohort size

> **HINT** *You are approximating a learning curve. Consider computing confidence intervals at each point using bootstrap resampling on the test set.*

✓ THE SHORT ANSWER

This exercise constructs a learning curve — a plot of model performance vs training set size. For most clinical ML models, AUROC grows steeply at small cohort sizes (200 – 2,000 patients), then slows and plateaus as you approach the full dataset. The plateau is described by the bias-variance tradeoff and, more formally, by learning theory's PAC (Probably Approximately Correct) framework. For a well-regularised logistic regression on this dataset, the plateau typically appears around 3,000 – 8,000 patients.

✓ CODE: GENERATING THE LEARNING CURVE

```
import numpy as np
from sklearn.utils import resample
from sklearn.metrics import roc_auc_score

SIZES   = [200, 500, 1000, 2000, 5000, 10000, 20000]
N_BOOT  = 200   # bootstrap samples per size for CI
results = []

for n in SIZES:
    # Sample n patients from training set (patient-level, stratified)
    pos_pids = [p for p in train_pids if sepsis_flag[p]==1][:int(n*0.07)]
    neg_pids = [p for p in train_pids if sepsis_flag[p]==0][:n-len(pos_pids)]
    sample_pids = pos_pids + neg_pids
    df_s = df_train[df_train['PatientID'].isin(sample_pids)]

    X_s = df_s[ALL_FEATURES].fillna(0).values
    y_s = df_s['SepsisLabel'].values
    X_s_sc = scaler.transform(X_s)

    lr_s = LogisticRegression(max_iter=500, class_weight='balanced', C=0.1)
    lr_s.fit(X_s_sc, y_s)
    probs = lr_s.predict_proba(X_test_sc)[:, 1]
    point_auroc = roc_auc_score(y_test, probs)

    # Bootstrap CI on test set
    boot_aurocs = []
    for _ in range(N_BOOT):
        idx = resample(range(len(y_test)), random_state=None)
        boot_aurocs.append(roc_auc_score(y_test[idx], probs[idx]))
    ci_lo, ci_hi = np.percentile(boot_aurocs, [2.5, 97.5])
    results.append({'n':n, 'auroc':point_auroc, 'lo':ci_lo, 'hi':ci_hi})
```

```
    print(f'n={n:>6}  AUROC={point_auroc:.4f}  95%CI [{ci_lo:.3f}, {ci_hi:.3f}]')
```

✓ WHAT TO EXPECT: ILLUSTRATIVE RESULTS

Training patients	Expected AUROC (approx)	95% CI width	Clinical interpretation
200	0.72 – 0.76	± 0.04 – 0.06	Far below clinical utility threshold; model cannot generalise reliably
500	0.76 – 0.80	± 0.03 – 0.05	Still high variance; wide CI means results may not replicate
1,000	0.79 – 0.83	± 0.02 – 0.04	Approaching useful territory; CI still wide relative to expected gains
2,000	0.81 – 0.84	± 0.02 – 0.03	Performance stabilising; CI beginning to tighten
5,000	0.83 – 0.86	± 0.01 – 0.02	Plateau region beginning; marginal gains from more data slowing
10,000	0.84 – 0.87	± 0.01	Near-plateau; adding 5,000 more patients yields < 0.5 AUROC points
20,000+	0.85 – 0.88	± 0.005 – 0.01	Plateau; architecture/features now matter more than data volume

✓ THE STATISTICAL CONCEPT: LEARNING CURVES & PAC THEORY

The learning curve describes the relationship between training set size and generalisation error. Three regions:

High-variance region (small n): The model has enough capacity to fit the training data but not enough data to estimate the decision boundary precisely. Training AUROC is high; test AUROC is much lower. More data helps dramatically.

Transition region (medium n): Training and test AUROC converge as the model sees enough diversity to generalise. Marginal returns per additional patient are decreasing.

Bias-limited region (large n): Model performance is now limited by its functional form (linear for LR) or feature quality, not by data volume. Adding more patients yields diminishing returns — the gap is now irreducible bias.

Formally, PAC theory states that with n training examples, a hypothesis class of VC dimension d achieves generalisation error bounded by $O(\sqrt{d/n})$. For logistic regression on 120 features, $d \approx 120$. The

bound becomes tight (error < 0.05) when n > ~48,000 — which means the full 32,000 training patients is near but not quite at the theoretical bound, explaining why the curve hasn't fully plateaued.

Practical design implication

If your hospital has only 500 ICU sepsis patients per year, a bespoke single-site model will be in the high-variance region for years. This is the primary argument for federated learning (training across multiple hospitals without sharing raw data) and for transfer learning (fine-tuning a model pre-trained on PhysioNet on local data). The learning curve quantifies exactly how much your single-site model suffers from insufficient data.

Q4 Temporal distribution shift: what breaks if Set A is from 2010 and Set B from 2020?

HINT *Think about changes in Sepsis-3 definition adoption, evolving ICU monitoring technology, and changes in treatment protocols (e.g. early goal-directed therapy fell out of favour around 2015).*

✓ THE SHORT ANSWER

At least five distinct mechanisms would invalidate a model trained on 2010 data and tested on 2020 data: (1) label definition change (Sepsis-2 → Sepsis-3 in 2016), (2) treatment protocol change (EGDT abandoned ~2014–2015), (3) monitoring technology change (new sensors, new lab assays), (4) patient population change (older patients, more comorbidities), and (5) care team behaviour change (earlier antibiotics culture after surviving sepsis campaigns). Each mechanism shifts the joint distribution P(X, Y) in a different way, breaking different model assumptions.

✓ THE FIVE MECHANISMS IN DETAIL

1. Label definition change: Sepsis-2 → Sepsis-3 (2016)

Pre-2016 label (Sepsis-2)	Two or more SIRS criteria (HR>90, RR>20, Temp<36 or >38, WBC<4 or >12) + suspected infection
Post-2016 label (Sepsis-3)	SOFA score increase ≥ 2 points + suspected infection
What changes	Sepsis-3 identifies a sicker population (organ dysfunction required). Some patients who were 'sepsis' under Sepsis-2 are 'infection without organ dysfunction' under Sepsis-3. The positive label shifts toward more severe cases.
How this breaks the model	A model trained on Sepsis-2 labels will learn to detect SIRS criteria (relatively mild physiological changes). A model validated on Sepsis-3 labels must detect organ dysfunction (more severe but different pattern). AUROC will drop because the model is predicting the wrong phenotype.

2. Treatment protocol change: EGDT abandoned (~2014–2015)

Pre-2015 treatment (EGDT)	Early Goal-Directed Therapy: aggressive fluid resuscitation targeting specific CVP, MAP, ScvO2 goals within 6 hours
Post-2015 treatment	ProCESS, ARISE, ProMISe trials showed EGDT no worse than usual care. Hospitals moved to more liberal, individualised fluid management
What changes in the data	Pre-2015 sepsis patients received large fluid boluses rapidly (often 3–6L crystalloid in 6 hours). Post-2015 patients received less fluid. CVP, MAP, and fluid balance trajectories shifted.
How this breaks the model	The model trained on pre-2015 data associates certain MAP/CVP patterns with treated sepsis. Post-2015 patients with the same underlying disease receive different treatment, producing different vital sign trajectories. The model's learned associations no longer hold.

3. Monitoring technology change: new sensors and lab assays

Pre-2015 ICU monitoring	Limited continuous monitoring; lactate measured intermittently; limited EtCO2 outside intubated patients
Post-2015 / 2020 monitoring	Continuous ScvO2 catheters, more frequent lactate measurement culture after Surviving Sepsis Campaign, improved ventilator monitoring
What changes in the data	Missingness patterns shift: lactate missingness may drop from 85% to 60% as it becomes standard of care. EtCO2 may become more common. New features may appear that the 2010 model never saw.
How this breaks the model	The model was trained assuming lactate is typically missing; high lactate availability signals that the patient is being resuscitated. This informative missingness signal reverses: now lactate is ordered routinely, so its presence no longer carries the same clinical signal. Imputation strategies built for 85% missingness are wrong at 60%.

4. Patient population change: aging, comorbidities, and polypharmacy

2010 ICU population	Younger average age; fewer immunosuppressed patients; less prevalent chronic kidney disease
2020 ICU population	Older average age; more cancer patients on chemotherapy; higher CKD prevalence (affects creatinine baseline); COVID-era changes from 2020 onward
What changes in the data	Baseline feature distributions shift (higher baseline creatinine, blunted tachycardia from beta-blockers, immunosuppression masking fever). The 'normal' range for many features shifts across the decade.
How this breaks the model	A creatinine of 1.8 mg/dL was alarming in 2010 (above normal); in 2020 a patient on long-term dialysis might have baseline creatinine of 5.0 mg/dL — so 1.8 is actually an improvement. The model has no concept of patient-level baselines, so it will systematically over-alert on patients with comorbid renal disease.

5. Care team behaviour change: earlier intervention culture

Pre-2010 clinical culture	Sepsis identified late in deterioration trajectory; antibiotics sometimes delayed hours pending cultures
Post-2015 / 2020 clinical culture	Hour-1 bundle mandated by Surviving Sepsis Campaign; many hospitals mandate blood cultures and antibiotics within 1 hour of suspected sepsis
What changes in the data	In 2020, antibiotics are given earlier in the clinical trajectory — sometimes before SOFA score reaches the labelling threshold. The temporal gap between physiological deterioration and the label narrows.
How this breaks the model	A 2010-trained model learned that the pre-label physiological window is 4–8 hours. In 2020, earlier intervention may have already altered vital signs before the label is assigned, compressing or eliminating the predictable pre-label window. The model expects a window that no longer exists.

✓ HOW TO DETECT AND MITIGATE TEMPORAL SHIFT

Detection method	What it finds	Mitigation	When to use
Covariate shift test (KS test per feature between years)	Features whose distributions changed significantly over time	Re-weight training examples or retrain on recent data	Before deployment or at regular retraining intervals
Label shift test (compare prevalence over time)	Changes in sepsis rate over time (reflects definition changes)	Recalibrate model output probabilities using recent label statistics	When upgrading from Sepsis-2 to Sepsis-3 labelling
Performance monitoring (AUROC decay over time)	Gradual model degradation in production	Trigger retraining when AUROC drops below threshold	Continuous production monitoring
Concept drift test (detect P(Y|X) change)	Changes in how features relate to outcome (treatment effect)	Add time as a feature; use time-aware ensemble	When treatment protocols change

The bottom line

If Set A is from 2010 and Set B from 2020, you should NOT train on A and test on B as if they are equivalent samples from the same population. They are not. The correct approach is: (1) explicitly test for distribution shift between the two sets before merging, (2) if shift is detected, weight training examples by inverse propensity of their time period, (3) consider a temporal split as your primary evaluation protocol rather than a random patient split, and (4) document the temporal scope of your training data as a fundamental limitation in any publication or deployment proposal.

Self-Assessment Checklist — Step 0

Check each item after completing Step 0. If you cannot check an item, re-read the relevant section above.

■ Environment & Setup

- ☐ I can explain why pinned versions matter and have created a requirements.txt
- ☐ My notebook runs end-to-end with fixed random seeds (np.random.seed(42) and torch.manual_seed(42))
- ☐ I can state the file count for Set A (20,336) and Set B (20,000) without looking it up
- ☐ I understand the .psv format and can identify the 4 column groups (vitals, labs, demographics, label)
- ☐ I have run the integrity check and confirmed both sets downloaded completely

■ Q1 — Privacy & Credentialing

- ☐ I can name at least 5 of HIPAA's 18 Safe Harbor identifiers from memory
- ☐ I can explain why date shifting alone does not make MIMIC open-access
- ☐ I can describe in one sentence why the 2019 Challenge dataset does not require CITI credentialing
- ☐ I understand what a Data Use Agreement (DUA) is and when one is required

■ Q2 — Production EHR Integration

- ☐ I can describe the difference between HL7 FHIR R4 and HL7 v2 in 2 sentences
- ☐ I can estimate end-to-end latency from nurse charting to model inference (5 – 75 min range)
- ☐ I understand why HIPAA BAA is required for cloud deployment and which vendors offer it
- ☐ I can sketch the 6-stage production pipeline (EHR → integration engine → buffer → feature service → model → alert)

■ Q3 — Learning Curves

- ☐ I can describe the three regions of a learning curve (high-variance, transition, bias-limited)
- ☐ I have run the learning curve code for at least 4 cohort sizes and recorded the AUROCs
- ☐ I can explain PAC theory in 2 sentences and connect it to the plateau I observe
- ☐ I understand why federated learning becomes relevant when single-site data is small

■ Q4 — Temporal Distribution Shift

- ☐ I can name 5 mechanisms by which 2010 → 2020 shift would invalidate the model
- ☐ I can explain the Sepsis-2 vs Sepsis-3 label difference and when it changed (2016)
- ☐ I can describe what EGDT was and why its abandonment shifts the data distribution
- ☐ I know at least 2 statistical tests to detect distribution shift before merging cohorts
- ☐ I can propose a mitigation strategy for each type of shift I identified

How to use this cheatsheet responsibly

This document contains full model answers. The deepest learning happens when you attempt each challenge question independently first — ideally with your notebook open — and use this as a check and enrichment resource afterward, not a shortcut. The challenge questions are designed to have no single right answer; the answers here represent one well-reasoned response, not the only valid one.